DTU Compute
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Machine Learning for Semi-Automated Annotations of the Linear A Inscriptions

Bachelor Thesis

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Preface

This Bachelor's thesis was prepared at the Department of Applied Mathematics and Computer Science at the Technical University of Denmark in fulfillment of the requirements for acquiring a Bachelor of Science in Engineering (BSc Eng) degree in General Engineering (Cyber Systems).

Kongens Lyngby, April 4, 2026

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Abstract

This thesis is in the workings... [Write something when finished with all thesis chapters]

Acknowledgements

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Generative AI tools were occasionally used as support for brainstorming, linguistic editing, code exploration, and research during the development of this project.

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Acronyms

HT Haghia Triada Tablet

SAM Segment Anything Model

mSAM MobileSAMv2

SA-1B Segment Anything 1 Billion

ViT Vision Transformer

CNN Convolutional Neural Network

NLP Natural Language Processing

LLM Large Language Model

MLP Multilayer Perceptron

Otsu Otsu's method

Gaussian Gaussian adaptive thresholding with bilateral pre-filtering

CannyFill Canny edge detection with morphological closing

GC+brush GrabCut with brush seeding

mSAM+pts MobileSAMv2 with automatic point prompts

YOLO You Only Look Once

Cefael Collections de l'École française d'Athènes en ligne

GORILA Godart and Olivier, Recueil des inscriptions en linéaire A

PDR Project Definition Report

IPR Intellectual Property Rights

GMM Gaussian Mixture Model

IoU Intersection over Union

Dice Dice-Sørensen coefficient

TPE Tree-structured Parzen Estimator

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CHAPTER 1

Introduction

1.1 Background

When the British archaeologist Arthur Evans in the beginning of the 20th century went to excavate Knossos on Crete, he found a huge amount of clay tablets written in a previously unknown linear script. Some documents were clearly of a newer script, which he called *Linear B*, while others were of an older one, which he named *Linear A*. Some fifty years later, the British amateur archaeologist Michael Ventris finally succeeded in deciphering the newer script Linear B, which he proved to be representing Mycenaean Greek, the earliest attested form of Greek. But the same hypothesis did not hold for the older script Linear A, whose encoded language remained unknown, still yet to be deciphered.

A key obstacle to its decipherment has been the small size of its corpus, lately consisting of only around 2,500 inscriptions (about 1,400 if only including those well-preserved enough to be readable) - significantly smaller than that of its younger counterpart that now spans more than 4,500 inscriptions (which are comparatively longer and better preserved).[1,2] But new inscriptions are still being discovered, with the corpus most recently extended in 2025 by over 100 new documents.[3,4]

Another challenge has been the lack of digital representations of these inscriptions, in a form suitable for statistical analysis. While photographs with accompanying drawings of nearly the entire corpus are available online through Collections de l'École française d'Athènes en ligne (Cefael) (see Godart and Olivier (1976–1985)), these are only available in print

[1] Salgarella 2020.
[2] Salgarella 2025, p. 13, 46.
[3] Del Freo and Zurbach 2025.
[4] Consiglio Nazionale delle Ricerche 2025.

form. This motivated the development of the palæographical[5] database *The Signs of Linear A* (SigLA), which seeks to "[develop] a systematic, exhaustive and user-friendly open access database of all Linear A inscriptions [...] in order to carry out statistical and palæographic analyses within the epigraphic corpus" (Salgarella and Castellan 2020). However, the last addition to SigLA was in 2021, and the project has so far managed to document just 772 of all currently known inscriptions.[6,7]

1.1.1 A problem of manual digitization

The drawings currently stored on SigLA were made by Ester Salgarella using the painting tool Krita. For each document, she manually created her own annotated drawings based on the originals from Cefael. From these, she proceeded to create layered Krita files with metadata for each sign, which could then at last be imported to SigLA.[8] According to her (personal communication), this has been a highly time-consuming undertaking. And the necessity of this kind of labor-intensive work has arguably been the biggest reason to why SigLA still does not contain the entire available corpus.

The aim of this project is therefore to investigate whether parts of this process can be automated. More specifically, whether some kind of semi-automatic segmentation tool can accelerate the workflow, while maintaining high-quality outputs that are still compatible with SigLA's database.

1.1.2 A problem of small data

The small size of the Linear A corpus and the irregularities in quality of its documents is a huge problem for the application of deep learning approaches, which typically require large amounts of data to perform well. Consequently, this project will instead seek to make use of generalized

[5]The term "palæographical" refers to the study of ancient writing systems.
[6]SigLA 2021a.
[7]SigLA 2021b.
[8]Salgarella and Castellan 2020.

machine learning and computer vision, in particular Meta's state-of-the-art interactive segmentation tool, the Segment Anything Model (SAM) (Kirillov et al. 2023), which is pre-trained on a massive and diverse dataset of over 1 billion masks. SAM's generalization and interactivity (allowing user-placed markers that guide the model) might thus allow for semi-automatic annotation of Linear A inscriptions without the need for large amounts of training data, which is unfortunately not available in this context.[9]

[9]By the "interactivity" of SAM, what is meant is NOT any built-in user interface in the form of a GUI, but rather the possibility of prompt-based placement of rectangles of focus, and points of additions and deletions that guide the model's prediction.

1.2 Related work

[THIS WHOLE SECTIONS NEEDS TO BE REWRITTEN, AS IT IS CURRENTLY JUST A VERY ROUGH DRAFT.]

This thesis places itself at the intersection of computer vision, digital humanities, and epigraphy (the study of inscriptions). Since it is a thesis of engineering, computer vision will remain the primary focus, both in terms of the research questions, and the methods used to answer them. As a result, the other two fields will merely provide the context and motivation for the research. The prior work and initial literature review in focus will thus concentrate on: 1. the relevant theory and state-of-the-art within computer vision (SAM, etc.), 2. the palæographical and digital humanities work that this project will seek to support (SigLA), and 3. the overall context thereof (Linear A).

1.2.1 Research context

For the context of Linear A and its corpus, the background section above can be referred to, citing the sources for: its size (Salgarella 2020) and (Salgarella 2025), its new addition (Del Freo and Zurbach 2025), and where it's accessed (Godart and Olivier 1976–1985). And for the epigraphical and digital humanities work, the background section can also be referred to, citing the sources for SigLA: its paper (Salgarella and Castellan 2020), its about page (SigLA 2021a), and its document repository (SigLA 2021b). Here, additional context can also be provided by Ester Salgarella herself, as she is a co-supervisor of this project, and the main person behind SigLA.

1.2.2 Computer vision literature

As for the relevant theory and state-of-the-art within computer vision, the literature there can be further split into three parts; 1. sources of general image processing, 2. sources concerning SAM and its implementation, and 3. related computer vision script segmentation work for comparison.

For image processing, a general theory book (Gonzalez and Woods 2018) can be cited, as well as more method-specific papers (when other methods are used) like Otsu thresholding (Otsu 1979), and of course the OpenCV Python documentation (OpenCV Team 2024) for short implementation reference. Then, for SAM, these are: its paper (Kirillov et al. 2023), its lighter implementation MobileSAM (C. Zhang et al. 2023) - and the newest version of that one MobileSAMv2 (Zhao et al. 2023). Lastly, for related segmentation work there exists a paper using deep learning (in contrast to the methodology of this thesis) obtaining results better than state-of-the-art (P. Zhang, Li, and Sun 2024), and a lighter EU-funded historical document segmentation tool *dhSegment* (Oliveira, Seguin, and Kaplan 2018) also using deep learning.

1.3 Problem statement

As previously stated, the aim of this project is to investigate whether parts of the process of creating annotated drawings for SigLA can be automated. More specifically, whether some kind of semi-automatic segmentation tool using interactive and generalized computer vision - specifically the Segment Anything Model (SAM) - can accelerate the workflow, while maintaining high-quality outputs that are still compatible with SigLA's database. But what would be the relative segmentation quality of such a tool? How automatic would it be exactly? And by how much could it improve the efficiency of the workflow? For a decomposition of the problem statement into its three research questions - each accompanied by their respective operationalization - see the following subsection:

1.3.1 Research questions

1. **Segmentation performance**: How well does a SAM-based segmentation approach perform quantitatively in terms of segmentation accuracy, compared to simple classical image processing baselines when applied to Linear A document images? [Operationalization: SAM-based masks will be compared to classical image-

processing baselines using overlap metrics (e.g. IoU/Dice) against the ground truth segmentations derived from SigLA's Krita files for a minimum of 25 documents.]

2. **Degree of automation**: To which extent can user interaction be reduced by a SAM-based semi-automatic segmentation workflow, while still maintaining high-quality outputs? [Operationalization: The number of user interactions (e.g. clicks) required to achieve a certain level of segmentation quality (e.g. IoU/Dice) will be measured against ground truth (SigLA's Krita files) for a minimum of 25 documents.]

3. **Workflow efficiency**: How much reduction in annotation time can be achieved using the proposed tool compared to fully manual digitization, under controlled experimental conditions? [Operationalization: The time it takes Ester Salgarella to annotate manually compared to being assisted by the proposed segmentation tool will be measured for a minimum of 5 documents.]

1.4 Impact – innovation and application

The work of this thesis presents itself as a first proof-of-concept (and a mapping of the difficulties involved) in using SAM and modern computer vision to accelerate the expansion of the SigLA database. Its main contribution is a prototype of a semi-automatic annotation tool, along with an evaluation framework that can be used to assess the performance of itself and any potential future versions. A future version of this tool could potentially also be applied to Linear B or segmentation of inscriptions in general, as it is the whole field of Aegean studies that seems to suffer from severe under-digitization of the primary evidence (which is a huge problem for the scientific investigations of scholars). Hopefully, the work will inspire further research and development within the field of digital epigraphy. Particularly in the application of generalized and interactive computer vision approaches such as SAM on Linear A, and perhaps even

in the broader context of use on small corpus ancient writing systems, where the lack of big data is ill fit for approaches using deep learning.

1.5 Overview of the thesis

The rest of this thesis is structured into five main chapters (Data, Methodology, Results, Discussion, Conclusion), all split up into sections concerning each of the three research questions.

CHAPTER 2

Data

2.1 Sources

The dataset used in this thesis was constructed from 30 images of raw Linear A inscriptions, and their corresponding ground truth annotations. Cefael (see Godart and Olivier 1976–1985) was chosen as the source for raw document images, since it is the only online and widely publicly available source of the printed five volume corpus - Godart and Olivier, Recueil des inscriptions en linéaire A (GORILA). And as all raw document images on the pages of this corpus are accompanied by annotated drawings, an argument could be made to make use of those as ground truth annotations. But annotations provided by Ester Salgarella from SigLA (see Salgarella and Castellan 2020) were chosen instead, as they are more up to date, and are already in a binary mask format that is suitable for comparison with segmentation masks.

What follows in this chapter is a detailed description of this dataset, documenting its characteristics, its construction process, and its limitations and biases.

2.2 Characteristics and construction

The dataset was made to consists of 30 chosen image pairs of raw documents and their corresponding ground truth annotations. No data split was made, as there otherwise would not be enough data for a meaningful evaluation. Thus, the dataset was used both for hyperparameter optimization and final evaluation of the models. Only inscriptions with clay as writing support were included, as they are the most common and representative of the Linear A corpus, and were the only ones for which

SigLA ground truth annotations were already available. The raw document images from Cefael are screenshots of grayscale images within the scanned printed edition of the corpus, and are thus not of the highest quality, being the only widely publicly available images of the Linear A inscriptions. For an example of a raw document image and its corresponding ground truth annotation, see Figure 2.1.

((a)) Raw image (HT7b) **((b))** Ground truth (HT7b)

Figure 2.1: Example of a raw inscription image and its corresponding ground truth annotation.

2.2.1 Difficulty classification

The dataset was categorized into three difficulty levels (easy, medium, hard) based on the visual quality of the raw images and the clarity of the inscriptions. This classification was performed manually by inspecting each image and considering factors such as contrast, noise, and the presence of tablet breakages or general damage to the inscriptions. The concrete criteria used to classify the difficulty sets were as follows:

- **Easy:** labeled as such for documents with clear ingravings, good contrast, minimal noise, and constant lighting.
- **Hard:** labeled to those with poor contrast, significant noise, and substantial damage making it hard even for the naked eye to read the inscriptions.
- **Medium:** labeled to all those inbetween the qualities of easy and hard.

For visual examples of images within each of these difficulties, see Figure 2.2.

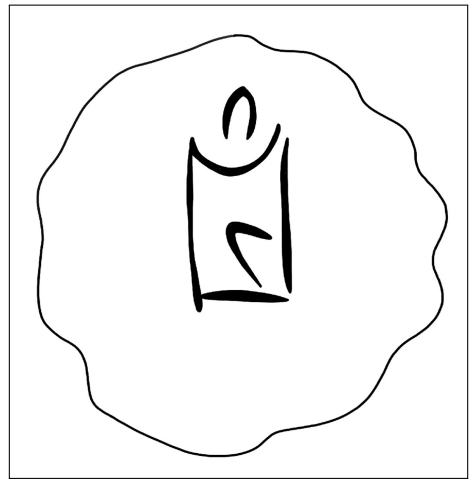
2.2.2 Image registration and preprocessing

To obtain any kind of meaningful comparison between model outputs and ground truth, the raw images and their ground truth annotations had to be aligned and registered to each other. This was done manually in Krita by downscaling the ground truth images to the individual scales of their raw counterparts, and aligning by translation and rotation, using the raw images as background reference.

2.3 Limitations and biases

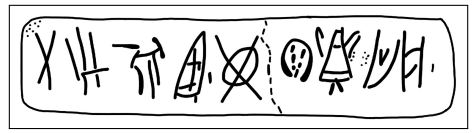
Even after image registration, ground truth images still did not perfectly align with the raw images, affecting the obtained segmentation quality. This misalignment is a source of noise in the evaluation.

((a)) Easy KHWc2104 (raw)

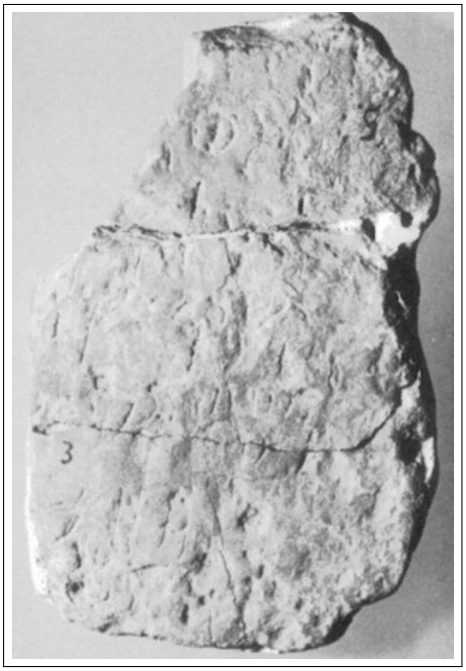


((b)) Easy KHWc2104 (gt)

((c)) Medium MA1a (raw)



((d)) Medium MA1a (gt)



((e)) Hard HT3 (raw)

((f)) Hard HT3 (ground truth)

Figure 2.2: Examples of images with different levels of segmentation difficulty: easy, medium, and hard. For each case, the raw image (left) and the corresponding ground truth annotation (right) are shown.

CHAPTER 3

Methodology

3.1 Common methodology

3.1.1 Development environment and tools

[TODO: EXTEND THIS SECTION] Python 3.10 was used for all code development (because it has such a big library of image processing functions and machine learning model implementations), with the following libraries:

- OpenCV for image processing and computer vision tasks.
- NumPy for numerical operations and array manipulations.
- Matplotlib for data visualization.
- *Add more!*

Krita was used for manual image registration, making sure the raw image ground truth pairs were properly aligned.

3.1.2 Common theory

[SHOULD THIS SECTION EXIST? IF SO, I SHOULD FINISH IT]

The basics of spatial filtering can be described by the following equation:

$$g(x, y) = T[f(x, y)]$$

, where $f(x, y)$ (also denoted as $I(x, y)$) is the input image, $g(x, y)$ is the output image.[1]

[1] Gonzalez and Woods 2018, p. 120.

3.1.2.1 Binary image thresholding

$$B(x,y) = \begin{cases} 1 & \text{if } I(x,y) > T \\ 0 & \text{otherwise} \end{cases}$$

[2]

3.1.2.2 Convolutional kernels

[3]

$$(w \star I)(x,y) = \sum_{i=-a}^{a} \sum_{j=-b}^{b} w(i,j) I(x-i, y-j)$$

where w is the convolution kernel (window) of size $m \times n$, and $a = (m-1)/2$ and $b = (n-1)/2$ are the half-width and half-height (assuming odd dimensions), respectively.[4]

3.2 Non-interactive segmentation ability

To answer the question of RQ1: *How does the initial segmentation quality of SAM compare to that of classic segmentation methods?*, the initial fully-automatic segmentation quality of a computationally efficient implementation of SAM (MobileSAMv2) was evaluated against a set of baseline models, using a suitable segmentation metric (i.e. Dice score) computed from the pairs of predicted and ground truth masks. All models were tuned by hyperparameter optimization to their best performance on the dataset, and the best performing version of SAM was compared against the best performing baseline model.

The following subsections therefore document the configuration of the baseline and state-of-the-art models, as well as the evaluation protocol used to answer RQ1.

[2] Gonzalez and Woods 2018, p. 743.
[3] Gonzalez and Woods 2018, p. 159.
[4] Gonzalez and Woods 2018, p. 157.

3.2.1 Baseline model configuration

The baseline models against which the best SAM implementation was evaluated (from simplest to most advanced) were:

- Global thresholding (Otsu's method)
- Local adaptive thresholding (Gaussian) with bilateral pre-filtering
- Edge-based segmentation (Canny edge detection with region filling)
- Energy-based segmentation (GrabCut with brushseeds)

Together, they cover a wide range of classic segmentation techniques that might be used for binary foreground extraction. What follows is a brief description of each of these methods, including their characteristics, how they work, and their implementation details.

3.2.1.1 Global thresholding (Otsu's method)

Otsu's method is a global thresholding method that works on gray-level histograms, which was originally proposed by Otsu (1979). As the simplest of the baselines, it represents the minimum acceptable segmentation quality, since it only takes the global gray level distribution into account, with no assumptions about any spatial relationships between pixels. It is therefore expected to perform poorly on images with erosion, uneven lighting, and differences in surface texture[5], which are the exact characteristics of the images to be segmented.

Here is how it works: First, let pixels of the raw image be divided into L shades of gray $[1, 2, ..., L]$. Then, let each pixel be assigned one of two classes C_0 or C_1, guided by a threshold T, where C_0 contains the shades $[1, 2, ..., T]$, and C_1 contains the rest $[T + 1, T + 2, ..., L]$. Otsu's method then works by performing an exhaustive search over all L gray levels to find an optimal threshold T^* that results in a maximum between-class variance σ_B^2:

$$T^* = \underset{1 \leq T < L}{\arg\max}\ \sigma_B^2(T)$$

[5] Gonzalez and Woods 2018, p. 751.

The between-class variance σ_B^2 is then defined by a function of the respective total probabilities of the two classes (ω_0 and ω_1) and the mean gray level value per class (μ_0 and μ_1):

$$\sigma_B^2(T) = \omega_0(T)\omega_1(T)[\mu_1(T) - \mu_0(T)]^2$$

3.2.1.2 Local adaptive thresholding (Gaussian) with bilateral pre-filtering

Local adaptive thresholding works by computing a threshold for each pixel based on its local neighborhood. This is an advantage in cases where the image has varying light conditions (which is the case for the images to be segmented), as the threshold adapts to local lighting.

Mathematically, the local threshold T is a function of a given pixel position (x, y) computed by a weighted sum of the neighborhood. The generated binary mask B is then defined as:[6]

$$B(x,y) = \begin{cases} 1 & \text{if } I(x,y) > T(x,y) \\ 0 & \text{otherwise} \end{cases}$$

To compute $T(x, y)$, there are a few weighting functions to choose from. For this baseline, the Gaussian kernel was used, as it can limit the influence of distant noisy pixels by assigning weights based on distance from the kernel center. With a Gaussian kernel G_σ, the threshold $T(x, y)$ is computed as a Gaussian-weighted sum of local intensities subtracted by a constant C. Using convolution ($\star$), this can be expressed as:[7]

$$T(x,y) = (G_\sigma \star I)(x,y) - C$$

In general, the weight at an offset (i, j) from the kernel center can be written as follows (where α is a normalization constant ensuring that the kernel sums to 1, and σ controls the spatial spread of the kernel):[8]

$$G_\sigma(i,j) = \alpha \cdot \exp\left(-\frac{i^2 + j^2}{2\sigma^2}\right)$$

[6] Gonzalez and Woods 2018, p. 761.
[7] OpenCV 2024.
[8] Gonzalez and Woods 2018, p. 167.

As a pre-filtering step, a bilateral filter (see Tomasi and Manduchi 1998) was applied to the input image before performing the Gaussian thresholding. In this way, more noise could be reduced while preserving inscriptions, as this filter is known to be effective at reducing noise while preserving edges. It takes three hyperparameters: the kernel window diameter d, the spatial standard deviation σ_{space}, and the intensity standard deviation σ_{color}. Discretely, its output intensity $I_{\text{bf}}(p)$ at a given pixel position p (where q is the position of one of its neighbors within the kernel window) can be defined as follows:[9]

$$I_{\text{bf}}(p) = \frac{\sum_q I(q)\, W_{\text{space}}(p,q)\, W_{\text{color}}(p,q)}{\sum_q W_{\text{space}}(p,q)\, W_{\text{color}}(p,q)}$$

Here, the spatial weight based on the distance between pixels p and q, and the color range weight based on the intensity difference between p and q are respectively given by:

$$W_{\text{space}}(p,q) = \exp\left(-\frac{\|p-q\|^2}{2\sigma_{\text{space}}^2}\right)$$

$$W_{\text{color}}(p,q) = \exp\left(-\frac{\|I(p)-I(q)\|^2}{2\sigma_{\text{color}}^2}\right)$$

It was decided to use the same σ value for both spatial and color weights, as this simplified the hyperparameter tuning (see Section 3.2.3.2) without significantly compromising the filter's performance. This gave in total 4 hyperparameters for this baseline (see Figure 3.1 for a list of these):

3.2.1.3 Edge-based segmentation (Canny edge detection with region filling)

Canny edge detection is a method that by itself only segments the edges of an image, excluding the enclosed regions within those edges. Here is how it works: First, the input image is smoothed using a Gaussian filter

[9] Pan 2025.

X

1. C: the constant subtracted from the Gaussian-weighted sum to compute the local threshold (see Section 3.2.1.2),
2. $d_{gaussian}$: the diameter of the kernel window for Gaussian adaptive thresholding (see Section 3.2.1.2),
3. $d_{bilateral}$: the diameter of the kernel window for bilateral pre-filtering (see Section 3.2.1.2),
4. σ: the spatial and intensity standard deviation for bilateral pre-filtering (see Section 3.2.1.2),

Figure 3.1: Hyperparameters for Gaussian adaptive thresholding with bilateral pre-filtering (Gaussian).

to reduce noise (using the Gaussian kernel as defined in the previous section):

$$I_s(x, y) = (G_\sigma \star I)(x, y)$$

Then, for the image gradient of each pixel, the magnitude $M_s(x, y)$ and the orientation $\theta_s(x, y)$ are computed:[10]

$$M_s(x, y) = \parallel \nabla I_s(x, y) \parallel \; = \sqrt{\left(\frac{\partial I_s}{\partial x}\right)^2 + \left(\frac{\partial I_s}{\partial y}\right)^2}$$

$$\theta_s(x, y) = \arctan\left(\frac{\partial I_s / \partial y}{\partial I_s / \partial x}\right)$$

Next, *non-maxima suppression*[11] is applied to retain only local maxima of $M_s(x, y)$ along the gradient direction $\theta_s(x, y)$, resulting in thinner edges. This is at last followed by *hysteresis thresholding*[12], classifying pixels as strong, weak, or non-edges with a low threshold T_l and a high threshold T_h, only preserving weak edges if connected to strong edges.

[10] Gonzalez and Woods 2018, p. 730.
[11] Gonzalez and Woods 2018, p. 730-731.
[12] Gonzalez and Woods 2018, p. 732.

In an attempt to fill the regions enclosed by the detected edges, morphological closing (dilation followed by erosion) was applied to the edge mask. This gave in total 2 hyperparameters (see Figure 3.2 for a list of these).

X

1. *sigma*: the constant used to determine the thresholds T_l and T_h for the hysteresis thresholding, where $T_l = \sigma \cdot \min(M_s)$ and $T_h = \sigma \cdot \max(M_s)$,

2. $d_{closing}$: the diameter of the kernel window for the morphological closing of the edges

Figure 3.2: Hyperparameters for Canny edge detection with morphological closing (CannyFill).

3.2.1.4 Energy-based segmentation (GrabCut with brushseeds)

GrabCut (Rother, Kolmogorov, and Blake 2004) is a classical interactive segmentation method, which makes for a good comparison to a deep learning interactive model like SAM. It is good at producing spatially coherent masks with little large-grained noise, which is ideal for the images to be segmented.

The algorithm is interactively initialized by the user, who traditionally provide a bounding box around the object of interest. Pixels outside this box are marked as definite background, while pixels inside it are marked as initially unknown. To further guide the segmentation, the user also has the option to provide an additional number of pixels as definite foreground ($B_n = 1$), and definite background ($B_n = 0$).

As an energy-based segmentation method, optimal segmentation is achieved by minimizing a cost (energy) function E that combines a data term D and a smoothness term S, given an input image I, a binary segmentation mask B, and a set of parameters θ:[13]

[13]This is a simplified version of the energy function that is used by GrabCut.

$$E(B, I, \theta) = D(B, I, \theta) + S(B, I)$$

The set of parameters θ then define two *Gaussian Mixture Models (GMMs)* (probabilistic models each consisting of a weighted sum of several Gaussian distributions) that over iterations will be tuned to model the foreground and background pixel distributions. These are used to compute the data term D whose energy contribution decreases the better B matches the GMMs of foreground and background. The smoothness term S then penalizes neighboring pixels that have different labels, encouraging spatial coherence in the segmentation.

At last, the optimal segmentation B^* is found iteratively, by alternating between optimizing the GMM parameters θ to model B, and minimizing the energy function with respect to B:

$$B^* = \arg\min_B E(B, I, \theta)$$

The automatic brush seeds given to GrabCut (see Figure 3.4 for a visual example) were derived from a Gaussian thresholded mask with bilateral pre-filtering (see Section 3.2.1.2). The foreground of the mask came to represent sure foreground pixels, while an eroded version of the background (the inverse of the mask) represented sure background pixels. A probable background region was then filled in taking up the remaining space created from the eroded background. This gave in total 6 hyperparameters (see Figure 3.3 for a list of these).

X

1. $d_{bilateral}$: the diameter of the kernel window for bilateral prefiltering (see Section 3.2.1.2),

2. σ: the spatial and intensity standard deviation for bilateral prefiltering (see Section 3.2.1.2)

3. C: the constant subtracted from the Gaussian-weighted sum to compute the local threshold (see Section 3.2.1.2),

4. $d_{gaussian}$: the diameter of the kernel window for the Gaussian adaptive thresholding (see Section 3.2.1.2),

5. $d_{prb\_erosion}$: the diameter of the kernel window for the erosion of the sure background, filled in by probable background,

6. $iters$: the number of iterations for GrabCut to run.

Figure 3.3: Hyperparameters for GrabCut with brush seeding (GC+brush).

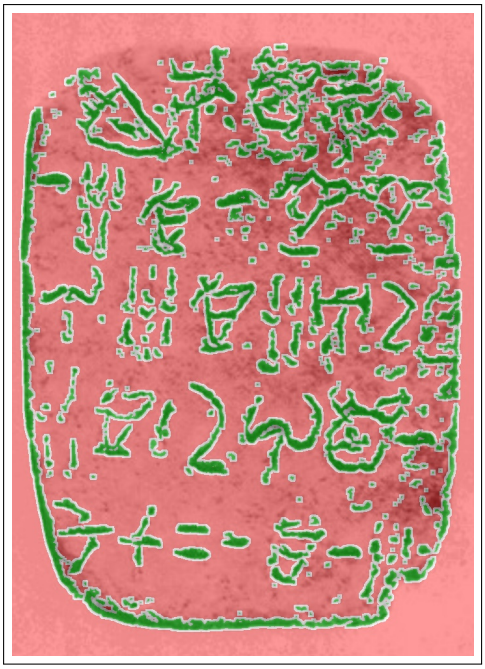


Figure 3.4: A visual example of seed distribution for GrabCut with brush seeding (GC+brush) on HT118. Green overlay represents sure foreground pixels, while red overlay represents sure background pixels, and white overlay represents probable background pixels.

3.2.2 State-of-the-art model configuration

To be compared against the classical baselines was the generalized deep learning Segment Anything Model (SAM) (see Kirillov et al. 2023). For the sake of computational efficiency, a mobile implementation of SAM - MobileSAMv2 (mSAM) (see Zhao et al. 2023) was used instead of the original. It is a lightweight version, which was designed to run more efficiently on mobile devices with a similar segmentation capability. More specifically, it is trained on a smaller sample of the same dataset of natural images as the original SAM; Segment Anything 1 Billion (SA-1B), and thus has roughly the same segmentation capabilities, although with a slightly lower performance.

As gray-scale images of clay tablets with Linear A inscriptions were not part of that dataset, mSAM had to be adapted to the specific characteristics of the domain. But as it could not be further tuned by training on the domain dataset (due to its small size), the only way to do so, was through the use of prompts. Specifically, automatic prompts, to make the non-interactive comparison to the baselines fair.

The main pipeline of mSAM thus ended up consisting of segmentation of the raw image by MobileSAMv2 into a binary mask, preceded by automatic prompt generation. What follows is therefore a brief description of the model architecture, and the automatic prompt generation method used to adapt it to the domain.

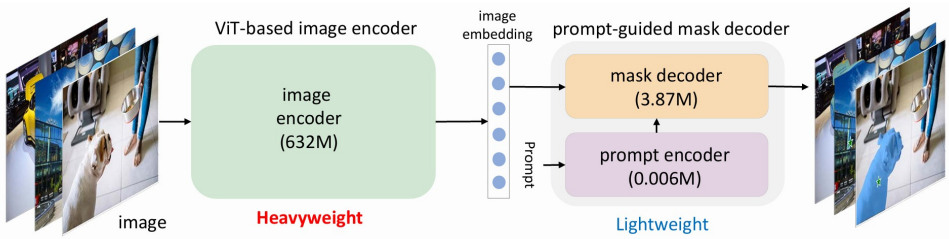


Figure 3.5: Architecture of SAM, from C. Zhang et al. (2023).

3.2.2.1 Model architecture

For both SAM and MobileSAMv2, the model architecture consists of an image encoder, and a prompt-guided mask decoder (see Figure 3.5). The image encoder used is a Vision Transformer (ViT)[14] whose purpose is similar to that of a Convolutional Neural Network (CNN), processing an input image to extract its features. But instead of convolutional layers, a ViT makes use of a transformer-based encoder traditionally used in Natural Language Processing (NLP) and Large Language Models (LLMs). As an alternative to words, it treats an image as a sequence of fixed size patches (regions of the image) embedded into vector representations with positional encodings, similar to how tokens work in NLP. These embedded patches are then processed through a repeated series of normalization, feed-forward Multilayer Perceptrons (MLPs), and in-between those, a mechanism known as self-attention that weights the patches according to their relevance to eachother.[15]

More specifically, given the input embedding X, self-attention computes the projections; quieries: Q, keys: K, and values: V, as products of X with respective weight matrices, and passes the information on as a new matrix (where d_k is the dimension of Q and K):

$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Here, the softmax function normalizes the output to a probability distribution (putting each element on the interval $]0, 1[$, with their sum being 1), which is then multiplied by V (again turned into logits / non-normalized probabilities).

Thanks to this attention mechanism, after a number of these series-layers, the ViT finally outputs an image embedding containing the spatial context of the image.[16] It is this image embedding, combined with embedded prompts from the prompt encoder (referred to as prompt to-

[14]The reason that MobileSAMv2 is computationally faster than the original is its lighter ViT implementation TinyViT (Wu et al. 2022) with only 5M parameters) compared to those of SAM (ViT-H, ViT-L, ViT-B with 632M, 307M, and 86M, respectively).

[15]see Vaswani et al. 2023, for the math behind *attention*.

[16]see Dosovitskiy et al. 2021, for how ViTs work in general.

kens) that are fed to the mask decoder to generate the final segmentation mask. However, before this happens, the prompt encoder takes any given sparse prompts (i.e. box or point prompts) as embeddings by summing their positional encodings with learned type-specifc embeddings (one foreground/background embedding per point prompt, and two opposite corner embeddings per box prompt).

These are then fed to the lightweight mask decoder (which is also transformer-based) along with the image-embedding from the ViT, and 3 learnable output token embeddings (see Figure 3.6). Self-attention is applied to the prompt and output tokens, and two-way cross-attention (same as self-attention but with one embedding as queries, and the other as keys and values, and vice-versa) is applied between the tokens and image embedding for them to influence and attend one another. To keep track of spatial coordinates and prompts, positional encodings and the original prompt embeddings are added as input at every attention layer.

At the end of this process, the image embedding is upsampled by convolution to the original image resolution. The three output tokens are then processed in parallel by two separate MLPs: one that generates their final segmentation masks by a dot product with the output and the upscaled image embedding, and another one that outputs a predicted IoU score for each mask. For single output masks, the chosen mask is the one with the highest of these associated scores.

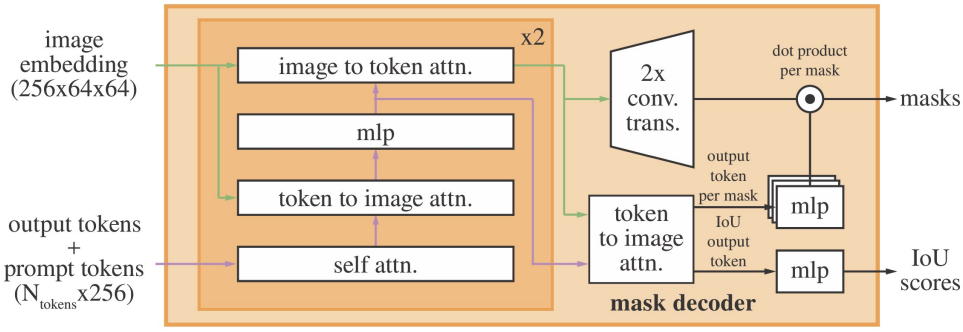


Figure 3.6: Architecture of SAM's mask decoder, from Kirillov et al. (2023).

3.2.2.2 Automatic prompting

Even though the MobileSAMv2 implementation already makes use of an automatic box prompt, it does so from a You Only Look Once (YOLO)[17] model trained on a subset of the same dataset as SAM, SA-1B.[18] Those box prompts are generalized for and work best on natural images. Thus, mSAM needed to be adapted through the use of other automatic prompts. But as it was particularly difficult to automatically create high quality box prompts without a trained object detection model, only point prompts were used for this adaptation. To perform a single ablation check of these automatic point prompts, two versions of MobileSAMv2 were then made for comparison:

- **mSAM**: MobileSAMv2 (with the original automatic box prompts from the YOLOv8 model),
- and **mSAM+pts**: MobileSAMv2 with automatic point prompts labeled as either foreground or background.

Just like in the case of GC+brush, the automatic point prompts for mSAM+pts (see Figure 3.8 for a visual example) were derived from a thresholded mask of the Gaussian adaptive thresholding baseline (see Section 3.2.1.2), This was done by randomly sampling a number of foreground and background pixels from the mask as prompts for mSAM+pts. Concretely, the foreground cloud of prompts were sampled from the foreground of the mask by n_{fgd} points, while the background cloud of prompts were sampled from the erosion of the inverse of the mask by n_{bgd} points. This gave in total 7 hyperparameters (see Figure 3.7) for mSAM+pts.

[17] Redmon et al. 2015.
[18] Zhao et al. 2023.

X

1. C: the constant subtracted from the Gaussian-weighted sum to compute the local threshold (see Section 3.2.1.2),
2. $d_{gaussian}$: the diameter of the kernel window for Gaussian adaptive thresholding (see Section 3.2.1.2),
3. $d_{bilateral}$: the diameter of the kernel window for bilateral pre-filtering (see Section 3.2.1.2),
4. σ: the spatial and intensity standard deviation for bilateral pre-filtering (see Section 3.2.1.2),
5. n_{fgd_points}: the number of foreground point prompts to be sampled from the foreground of the first Gaussian thresholded mask,
6. n_{bgd_points}: the number of background point prompts to be sampled from the sure background region,
7. $d_{gap\_erosion}$: the kernel size for erosion used to create a gap between the foreground and background regions.

Figure 3.7: Hyperparameters for MobileSAMv2 with automatic point prompts (mSAM+pts)

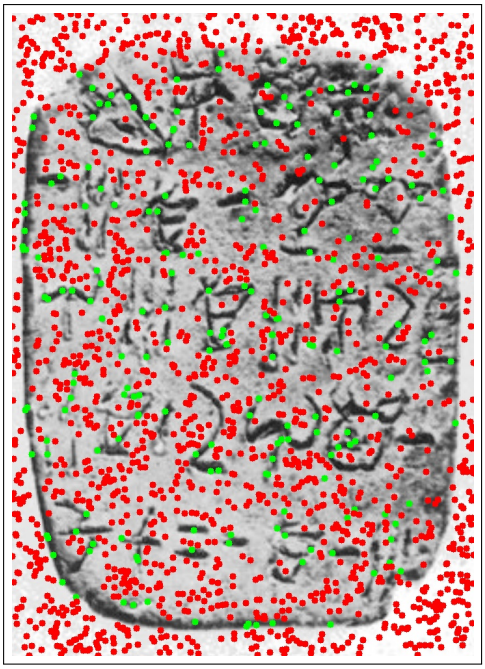


Figure 3.8: A visual example of prompt distribution for MobileSAMv2 with automatic point prompts (mSAM+pts), overlayed on HT118. Green points represent foreground point prompts, while red points represent background point prompts.

3.2.3 Evaluation protocol and performance metrics

For an evaluation of initial segmentation quality, relevant metrics and statistical tests needed to be defined to determine whether one model performed significantly better than another. Furthermore, all models with hyperparameters needed to be optimized to their best performance on the dataset, to ensure a fair comparison. Naturally, it was of particular interest to compare the best performing baseline model against the best performing version of MobileSAMv2.

What follows is therefore a description of the choice of metrics, the optimization of model hyperparameters, and the statistical tests for the evaluation, used to answer RQ1: *How does the initial segmentation quality of SAM compare to that of classic segmentation methods?*.

3.2.3.1 Choice of metrics

When it comes to evaluation of segmentation models, the two most commonly used metrics are: 1. the Dice-Sørensen coefficient (Dice) and 2. the Intersection over Union (IoU). The Dice score is most commonly used in cases where there are heavy class imbalances (e.g. lots of background compared to regions segmented), and when foreground objects are small and thin. It is naturally more foregiving than the related segmentation metric IoU, which instead tends to be rather strict, and is specifically used to evaluate segmentation results where small displacements are critical for results. Therefore, segmentation performance was reported using Dice. Mathematically, it is defined as twice the size of the intersection of the predicted segmentation mask $\hat{Y}$ and the ground truth mask Y, divided by their summed size:[19]

$$\text{Dice}(\hat{Y}, Y) = \frac{2 \cdot |\hat{Y} \cap Y|}{|\hat{Y}| + |Y|}$$

Thus, the raw metrics ended up consisting of the Dice score for each image, and each model.

[19] Mohammadi, Mollazade, and Behroozi-Khazaei 2025.

3.2.3.2 Hyperparameter optimization

To ensure a fair comparison between the models, all models with hyperparameters were tuned to their best performance on the dataset using Optuna, a hyperparameter optimization framework. The optimization itself was performed using the Tree-structured Parzen Estimator (TPE) algorithm (see Bergstra, Yamins, and Cox 2013)[20], which is a Bayesian optimization method that models the relationship between hyperparameters and performance to efficiently explore the hyperparameter space and find the optimal ones. Hence, it is more sophisticated than a random search, which just samples different hyperparameter combinations without considering the results of past evaluations.

It works by building two probabilistic models: one for the hyperparameter configurations that have resulted in good performance, and another for those that have resulted in poor performance. Then, it uses these models to select the next hyperparameter configuration to evaluate, by maximizing the expected improvement over the current best performance. This way, it can focus the search on promising regions of the space of possible hyperparameters without having to rely on chance or exhaustive search.

3.2.3.3 Statistical tests

To determine whether the observed differences in scores between models were statistically significant, paired t-tests were conducted. The paired t-test has two main assumptions: 1. independence of observations, and 2. normality of residuals. The first assumption is already satisfied a priori, since the segmentation performance of one image does not directly influence the performance on another. However, the second assumption depends on results. Thus, the residuals of the paired differences in Dice scores between the two models in question needed to be checked for normality using a QQ-plot and the Shapiro-Wilk test. And in case the normality assumption was violated, the non-parametric domain-specific Wilcoxon signed-rank test would be used instead.[21]

[20] also see Bergstra, Bardenet, and Kégl December 2011.
[21] Rainio, Teuho, and Klén 2024.

3.3 Interactive segmentation ability

Here, the interactivity and quality of human-guided SAM (and GrabCut for comparison?) will be evaluated against the initial segmentation results from Section 3.2. [Add some context of RQ2 here, and how the question will be operationalized in bigger picture (though more detail than in introduction).]

3.3.1 Interactive tool architecture

3.3.2 Experimental design for automation evaluation

3.3.3 Definition of automation metrics

3.4 Workflow efficiency

[Add some context of RQ3 here, and how the question will be operationalized in bigger picture (though more detail than in introduction).]

3.4.1 Export to Krita

[It is indeed possible via the Krita Python API to export the predicted segmentation masks as Krita layers, which can then be further edited by hand by a user. The question is only if I want to do so via the API (requires Krita installed), or via a Python export script. Krita's file format is open, so both are possible.]

3.4.2 Definition of workflow efficiency metrics

3.4.3 Experimental design of the user study

3.4.4 Analysis of results

CHAPTER 4

Results and Discussion

4.1 Non-interactive segmentation performance

After running the non-interactive segmentation evaluation of the models on the dataset, it was clear that mSAM did not perform better than the classical baselines (see Figure 4.1 for a plot of the results). The mSAM model with the in-built YOLO box prompts performed particularly poorly, being the worst of all with a mean Dice score of around 0.028. At the same time, mSAM+pts with the engineered point prompt clouds performed considerably better. With a mean Dice score of around 0.164, it performed better than Otsu and CannyFill (with mean Dice scores of 0.091 and 0.105, respectively), and comparably to the best performing classical baselines, Gaussian and GC+brush that each yielded Dice scores of 0.176 and 0.178, respectively.

The following subsections go into a bit more detail on the explanation of these results, the statistical significance of the differences in performance between mSAM and the best classical baseline (see Section A.1 in the appendix for the full results table), and a discussion of the fairness of this experiment.

4.1.1 Model comparison

Knowing the overall quantitative results (see Figure 4.1), it pays to look at a visual example to see what is going on behind the scenes. Looking at Figure 4.2 of the output masks per model from segmentation on the easy-difficulty HT118, it is clear that the Gaussian baseline manages

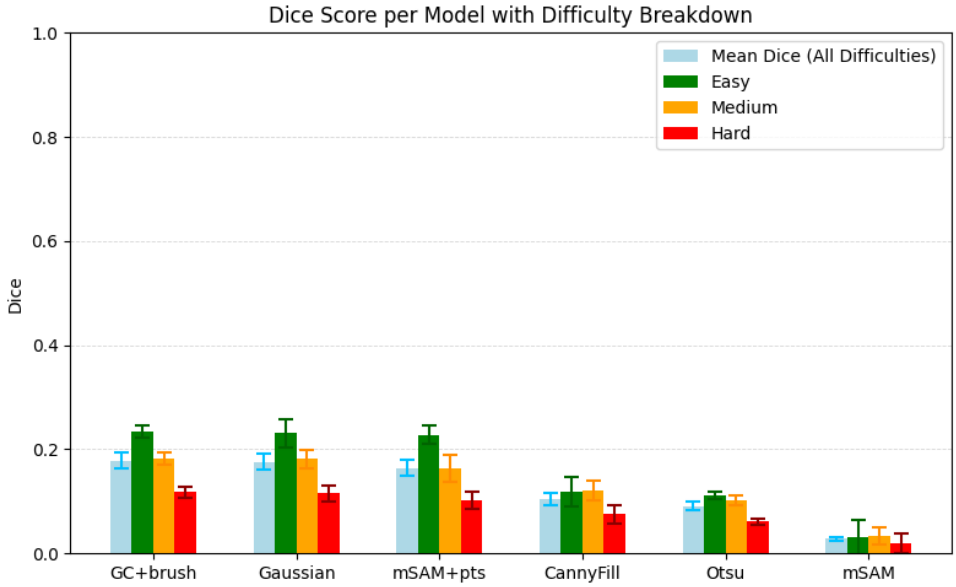


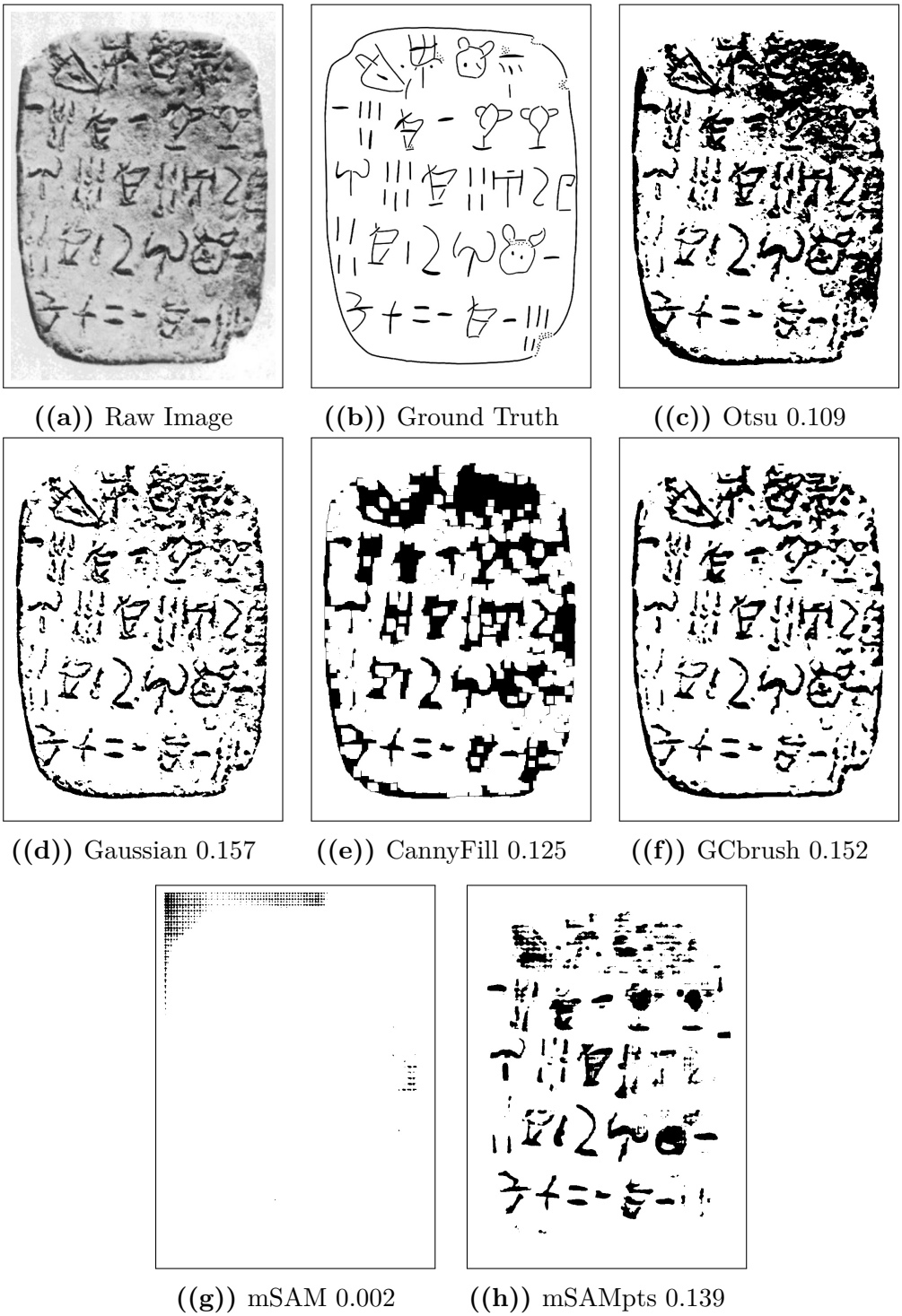
Figure 4.1: Dice Score performance per model. Split by difficulty. With error bars representing the standard error of the mean. Ordered descendingly from the left by mean performance.

to capture the information of this tablet the best[1], although with some noise. In this instance, GC+brush performs decently as well, but oversegments in a way that makes strokes a bit thicker than they should be. It compensates for this, however, by reducing noise. This is due to a combination of its probable background regions, which risk dilating the segmentation a bit, and its smoothness term (see Section 3.2.1.4), which simultaneously encourages spatially coherent masks.

The Otsu and CannyFill baselines perform a bit worse, although they capture almost all signs. Otsu also segments dark background regions as foreground, as they count as such according to its global threshold, turning a lot of its mask into noise. CannyFill also oversegments, due to its large kernel window size for morphological closing, which was necessary to fill in the gaps between edges, but gives a blocky look, and merges all nearby edges together.

[1]That tablet (HT118) is coincidentally also one of the instances, where Gaussian performs best of all.

The unprompted mSAM segments only background noise, producing nothing of value. However, the prompted mSAM+pts does produce a somewhat coherent mask, capturing almost every sign with less over-segmented areas than Otsu and CannyFill. Still, it oversegments many signs (especially those with background-space supposed to be inside of them), fades out signs where they are supposed to continue (due to loss of attention, as segmenting many separated signs at once is not what its attention weights have been trained for), and is not able to capture the outline of the tablet. But the signs that it does capture well, are captured with a good level of spatial coherence. And there is way less noise between them compared to the masks of the baselines, which is arguably a positive sign for the potential of SAM to perform well on this task once properly fine-tuned.



((a)) Raw Image ((b)) Ground Truth ((c)) Otsu 0.109

((d)) Gaussian 0.157 ((e)) CannyFill 0.125 ((f)) GCbrush 0.152

((g)) mSAM 0.002 ((h)) mSAMpts 0.139

Figure 4.2: Visual comparison of output masks per model from segmentation on HT118 (easy), shown with Dice scores next to the raw image and ground truth.

4.1.2 Statistical significance

As it was clear from the results, mSAM+pts did not perform better than the best performing classical baseline. To determine whether it performed significantly worse, a paired t-test was conducted, as residuals of the differences in Dice scores between GC+brush and mSAM+pts were approximately normally distributed. What follows is the documentation of that normality test, the paired t-test, and the results of both.

4.1.2.1 Normality of residuals

As the QQplot in Figure 4.3 shows, the residuals of the paired differences in Dice scores between GC+brush and mSAM+pts appear to be approximately normally distributed. Furthermore, the Shapiro-Wilk test for normality on these residuals yielded a p-value of $0.178 > \alpha = 0.05$, indicating that we fail to reject the null hypothesis of normality. Thus, the normality assumption for the paired t-test is satisfied, and its results can be considered valid.

4.1.2.2 Paired t-test

The paired t-test was conducted on the Dice scores obtained for each of the 30 images across GC+brush and mSAM+pts. The null hypothesis (H_0) stated that there is no difference in mean Dice scores between the two models, while the alternative hypothesis (H_1) stated that GC+brush has a higher mean Dice score than mSAM+pts.

Performing the paired t-test yielded a test statistic of $t \approx 2.656$ and a p-value of $P(T > t) \approx 0.013$. Thus, as $0.013 < \alpha = 0.05$, the null hypothesis could be rejected, and it could be concluded that GC+brush significantly outperforms mSAM+pts in terms of Dice score. Hence, it can be said that mSAM performed significantly worse in comparison to the best classical segmentation methods for non-interactive segmentation of Linear A tablets, even when automatically prompted.

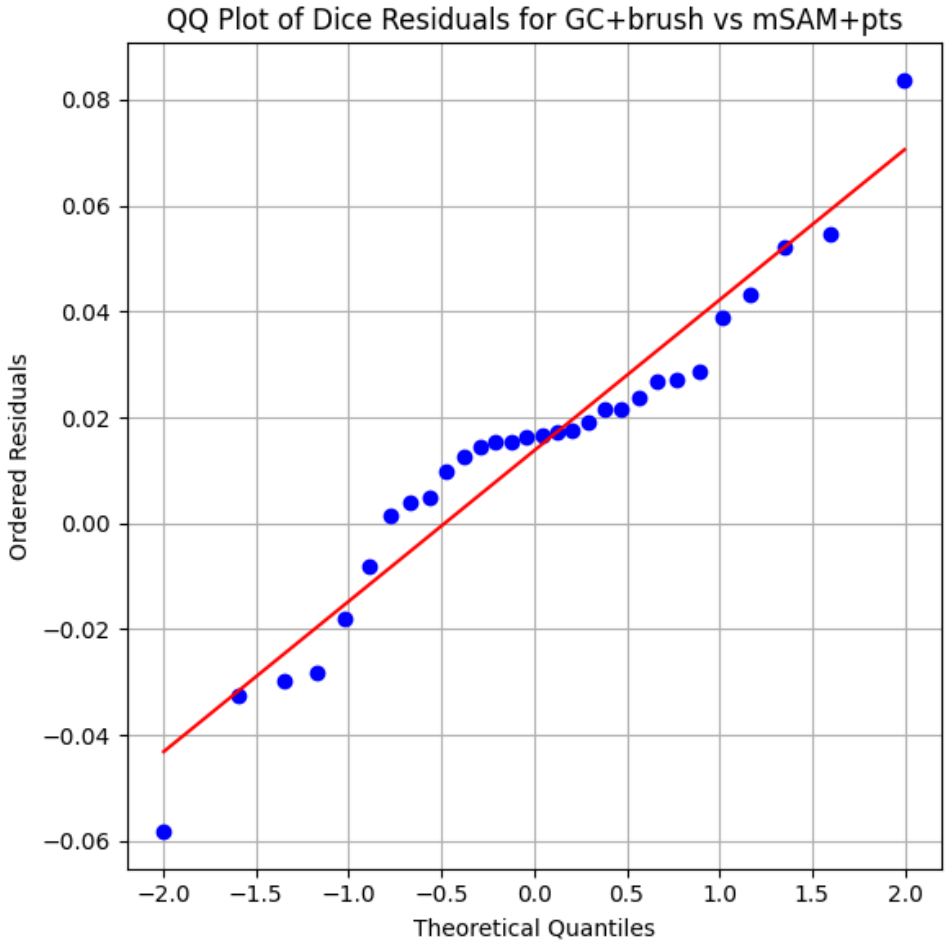


Figure 4.3: QQplot of the residuals of the paired t-test comparing GC+brush and mSAM+pts. The residuals appear to be approximately normally distributed, which supports the validity of the paired t-test results. Furthermore, the Shapiro-Wilk test for normality on these residuals yielded a p-value of $0.178 > \alpha = 0.05$, indicating that we fail to reject the null hypothesis of normality.

4.1.3 Fairness of the experiment

4.1.3.1 Data

The dataset used for evaluation was the same as the one used for hyperparameter optimization of models, which could be seen as a potential source of bias. Additionally, the dataset of 30 images is quite small, which makes it more susceptible to noise and outliers, although sufficiently sized to support the statistical analysis. Moreover, the ground truth annotations of the dataset were heavily biased, even as they were manually created and registered to the raw images. Therefore, Dice scores obtained from the evaluation were unreasonably low, and the performance of the models is not properly reflected by these scores. However, for comparison between models, the relative differences in Dice scores were still meaningful, as all models were evaluated on the same imperfectly aligned dataset.

4.1.3.2 Model implementation

In the case of the fairness of the SAM implementation chosen, the experiment only evaluates the performance of an untuned version. This implementation had most likely never seen Linear A tablet images during training, as it was pretrained by default on the large and diverse dataset of natural images SA-1B. Meanwhile, studies like Wang et al. 2025 have shown that the performance of SAM can be significantly improved by fine-tuning the model on a dataset that is more similar to the target domain. Furthermore, the model implementation used makes use of MobileSAMv2 with its small ViT image encoder TinyVit (Wu et al. 2022), which can compromise on segmentation quality. Thus, it does not come as a surprise that mSAM underperforms in comparison to the classical baselines. These are unlike SAM not restricted to natural images, and can therefore arguably be more easily adapted to the domain of Linear A tablets, when deep learning fine-tuning of SAM's weights is not an option. The results therefore suggest a need for a computationally efficient SAM implementation to be tuned to this domain, as the default model is not sufficient enough to prove its potential capabilities there.

4.2 Interactive segmentation performance

4.3 Workflow impact

CHAPTER 5

Conclusion

5.1 Summary of findings per research question

5.1.1 RQ1: Segmentation performance

5.1.2 RQ2: Degree of automation

5.1.3 RQ3: Workflow efficiency

5.2 Overall contribution

5.3 Future Work

5.3.1 Methodological extensions

5.3.2 Dataset expansion and diversification

5.3.3 Further automation opportunities

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APPENDIX A

An Appendix

A.1 Data from the non-interactive segmentation evaluation

Table A.1: Raw Dice score data on easy images. Label column represents the name of the document segmented, while all other columns represent Dice scores obtained for those documents for a given model.

Label	CannyFill	GC+brush	Gaussian	Otsu	mSAM	mSAM+pts
HT10a	0.179	0.255	0.254	0.131	0.057	0.233
HT118	0.125	0.152	0.157	0.109	0.002	0.139
HT12	0.148	0.27	0.261	0.192	0.048	0.218
HT17	0.158	0.233	0.225	0.071	0.03	0.263
HT22	0.024	0.192	0.191	0.136	0.057	0.163
HT26a	0.14	0.263	0.264	0.116	0.001	0.242
HT7a	0.115	0.126	0.133	0.116	0.02	0.159
HT7b	0.168	0.177	0.168	0.074	0.026	0.15
KHWc2104	0.005	0.438	0.439	0.069	0.027	0.496
PH2	0.12	0.235	0.225	0.096	0.046	0.218

Table A.2: Raw Dice score data on medium images. Label column represents the name of the document segmented, while all other columns represent Dice scores obtained for those documents for a given model.

Label	CannyFill	GC+brush	Gaussian	Otsu	mSAM	mSAM+pts
HT154B	0	0.138	0.152	0.073	0.032	0.122
HT16	0.162	0.175	0.167	0.081	0.003	0.183
HT2	0.071	0.107	0.117	0.094	0.021	0.103
HT4	0.11	0.237	0.24	0.157	0.061	0.22
HT40	0.122	0.154	0.161	0.088	0.029	0.126
HT6b	0.122	0.19	0.178	0.075	0.004	0.166
HT85b	0.124	0.111	0.104	0.06	0.035	0.096
HT8b	0.237	0.28	0.283	0.11	0.052	0.196
HT90	0.018	0.171	0.173	0.14	0.012	0.17
MA1a	0.238	0.259	0.243	0.144	0.09	0.25

Table A.3: Raw Dice score data on hard images. Label column represents the name of the document segmented, while all other columns represent Dice scores obtained for those documents for a given model.

Label	CannyFill	GC+brush	Gaussian	Otsu	mSAM	mSAM+pts
HT1	0.128	0.129	0.114	0.043	0.049	0.09
HT3	0.079	0.086	0.085	0.058	0.046	0.067
HT30	0.098	0.102	0.1	0.054	0.006	0.097
HT33	0.035	0.05	0.049	0.016	0	0.035
HT52a	0.002	0.093	0.098	0.065	0.006	0.076
HT85a	0.115	0.099	0.1	0.057	0.006	0.128
HT8a	0.177	0.211	0.203	0.061	0.041	0.157
HT94b	0.064	0.074	0.078	0.038	0.017	0.06
HT9a	0.021	0.217	0.208	0.147	0.02	0.235
HT9b	0.032	0.12	0.116	0.071	0	0.077

Table A.4: Summary Statistics per Model of Dice Score on all images.

Model	Mean Dice	Standard deviation	Standard error	n
CannyFill	0.105	0.066	0.012	30
GC+brush	0.178	0.082	0.015	30
Gaussian	0.176	0.08	0.015	30
Otsu	0.091	0.041	0.007	30
mSAM	0.028	0.023	0.004	30
mSAM+pts	0.164	0.089	0.016	30

Table A.5: Summary Statistics per Model of Dice Score on easy images.

Model	Mean Dice	Standard deviation	Standard error	n
CannyFill	0.118	0.058	0.018	10
GC+brush	0.234	0.087	0.027	10
Gaussian	0.232	0.086	0.027	10
Otsu	0.111	0.038	0.012	10
mSAM	0.031	0.02	0.006	10
mSAM+pts	0.228	0.104	0.033	10

Table A.6: Summary Statistics per Model of Dice Score on medium images.

Model	Mean Dice	Standard deviation	Standard error	n
CannyFill	0.12	0.079	0.025	10
GC+brush	0.182	0.06	0.019	10
Gaussian	0.182	0.057	0.018	10
Otsu	0.102	0.034	0.011	10
mSAM	0.034	0.028	0.009	10
mSAM+pts	0.163	0.051	0.016	10

Table A.7: Summary Statistics per Model of Dice Score on hard images.

Model	Mean Dice	Standard deviation	Standard error	n
CannyFill	0.075	0.055	0.017	10
GC+brush	0.118	0.055	0.017	10
Gaussian	0.115	0.051	0.016	10
Otsu	0.061	0.034	0.011	10
mSAM	0.019	0.019	0.006	10
mSAM+pts	0.102	0.058	0.018	10

$\sqrt{17}$
$\Omega \int \delta e^{i\pi} = -1$
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χ^2
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Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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